

quantum Equation-of-Motion (qEOM) protocol leveraging the computation of excited states [10, 11, 12], variations of the Lanczos algorithm [13, 14] and the continued fraction representations of dynamical correlation functions [15]. In the time domain, a standard approach is to use a suitable version of the well-known ancilla-based Hadamard test [16, 17], which requires long-range controlled quantum operations. While measurements of out-of-time-ordered correlators have been demonstrated on analog quantum simulators [18], large-scale simulations on digital platforms have remained limited, particularly on architectures with restricted qubit-qubit connectivity. Proof-of-principle demonstrations on digital devices include examples based on real-time dynamics [19, 16, 20], imaginary-time dynamics [6, 21] and variational approaches [22, 23, 24].

Alternative strategies to access dynamical susceptibilities without requiring ancilla qubits have been explored [25, 26], and, in particular, methods inspired by Linear Response (LR) experiments have been recently proposed [27, 28, 29]. Building on these ideas, here we present a comprehensive framework for computing RGFs on quantum processors, and construct a novel class of quantum algorithms which are already suitable for experiments on noisy devices. More specifically, we develop and extend the theoretical analysis of previous studies to make explicit the connection between LR functions and derivatives of parametrized quantum circuits. Building on this formulation, we introduce circuit constructions that allow RGFs to be accessed through circuit differentiation, including implementations based on parallel-in-time perturbations and stochastic estimation techniques, illustrated in Fig. 1. Within this framework, time-dependent correlation functions can be reconstructed from a single parametrized circuit family, enabling the extraction of dynamical structure factors and related spectroscopic quantities.

The manuscript is organized as follows. In Section 2, we provide an overview of the theoretical framework of LR theory, laying the foundation for our approach and establishing a unified formalism. In Section 3, we develop a general circuit-differentiation framework for computing RGFs and discuss multiple estimator constructions within it, including deterministic and stochastic approaches relevant to near-term devices and extensible to fault-tolerant settings. In Section 4, we review the models and analytical expressions used to illustrate our techniques, and in Section 5 we present and analyze the corresponding simulation results. Finally, in Section 6, we offer concluding remarks. Although we focus on reference toy models as examples, our results remain general and do not rely on any specific physical system. Throughout this work,  $\hbar$  is set to 1.

## 2 Background

Let  $H_0$  be the Hamiltonian describing a system in isolation and  $H = H_0 + H_s(t)$  be the total Hamiltonian for the system weakly coupled to an external time-dependent perturbation (or *source*)  $H_s(t)$  that can be turned on and off adiabatically. Let us consider the expectation value  $\langle o_{\mathbf{R}} \rangle = \langle \psi_0 | o_{\mathbf{R}} | \psi_0 \rangle$  of a local observable  $o_{\mathbf{R}}$ , evaluated on the ground state  $|\psi_0\rangle$  of  $H_0$ . Here  $\mathbf{R}$  indicates a generic 3D coordinate in the system and could be, for instance, a specific lattice site if the model is discretized. The source Hamiltonian can be further specified as

$$H_s(t) = \int s(\mathbf{r}', t') o_{\mathbf{r}'}(t') d\mathbf{r}', \quad (2)$$

assuming that the external perturbation couples linearly to the system operators  $o_{\mathbf{r}'}(t')$  at positions  $\mathbf{r}'$  through the force function  $s(\mathbf{r}', t')$ . Moving into the interaction picture, the operator  $o_{\mathbf{R}}$  evolves under the action of the unperturbed Hamiltonian  $H_0$  according to  $o_{\mathbf{R}}^I(t) = e^{iH_0 t} o_{\mathbf{R}} e^{-iH_0 t}$ . Meanwhile, the propagator for quantum states is given by the time-ordered exponential

$$U_s(t, t_0) = \mathcal{T} \exp \left( -i \int_{t_0}^t H_s^I(t') dt' \right), \quad (3)$$

where the interaction-picture source term is defined as  $H_s^I(t') = e^{iH_0 t'} H_s(t') e^{-iH_0 t'}$ .

The evolution in time of  $\langle o_{\mathbf{R}} \rangle$  induced by the action of the source can then be written as

$$\langle o_{\mathbf{R}}(t) \rangle_s = \langle \psi_0 | U_s^\dagger(t, t_0) o_{\mathbf{R}}^I(t) U_s(t, t_0) | \psi_0 \rangle. \quad (4)$$

Expectation values are from now on taken with respect to the ground state  $|\psi_0\rangle$ , unless otherwise stated. By expanding the propagator to *linear* order in the perturbation  $H_s$ , the change in expectation value of Eq. (4) up to a final time  $T$  is given by the Kubo formula [30]

$$\Delta \langle o_{\mathbf{R}}(T) \rangle_s = i \int_{-\infty}^T \langle [H_s^I(t'), o_{\mathbf{R}}^I(T)] \rangle dt', \quad (5)$$

where  $\Delta \langle o_{\mathbf{R}}(T) \rangle_s = \langle o_{\mathbf{R}}(T) \rangle_s - \langle o_{\mathbf{R}} \rangle$  denotes the difference between the perturbed and unperturbed expectation values and we introduced the usual commutator  $[A, B] = AB - BA$ . The integral is extended to  $t_0 \rightarrow -\infty$ , assuming that the perturbation is simply turned off in the distant past. Eq. (5) fulfills causality, as any change in the expectation value of the observable at time  $T$  can only be induced by a perturbation applied at a time  $t' < T$ . Additionally, if the observable  $o_{\mathbf{R}}$  is normal ordered with respect to the ground state  $|\psi_0\rangle$ , implying that  $\langle o_{\mathbf{R}} \rangle = 0$ , the first-order fluctuations from Eq. (5) simply correspond to the time-evolved expectation value  $\Delta \langle o_{\mathbf{R}}(T) \rangle_s = \langle o_{\mathbf{R}}(T) \rangle_s$ . Using Eq. (2), Eq. (5) can then be rewritten as

$$\langle o_{\mathbf{R}}(T) \rangle_s = \int d\mathbf{r}' \int_{-\infty}^T \mathcal{G}(\mathbf{R}, \mathbf{r}', T, t') s(\mathbf{r}', t') dt', \quad (6)$$